

Chapter 4

IRAS ASTEROID DATA PROCESSING

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This chapter describes the asteroid data processing from its beginnings in the IRAS Scientific Data Analysis Subsystem through the final products generated by the IRAS Minor Planet Survey. The former system contained two beginning points for asteroid processing: prediction of known-asteroid detections and accumulation of IRAS detections that might turn out to be asteroid sightings, known or unknown. The predictions provided information on processing quality, and the accumulation provided the large data set for ongoing asteroid identification. The latter system took advantage of this knowledge to refine the information into a set of products whose scope included asteroids not yet discovered at the time the data sets were created.

The scientific processing of the asteroid data began in the Science Data Analysis Subsystem (SDAS). This system reduced the observations to physical units and generated maps and catalogs of the fixed sources seen on the sky. SDAS was designed to recognize known solar-system sources and to save all other likely solar-system objects for later analysis. This analysis was accomplished using specially designed software known as the IRAS Minor Planet Survey (IMPS), which is described in §4.3 below.

The first step in recognizing asteroids in SDAS was the computation of their expected apparent positions on the sky. This task was carried out by the Asteroid Predictor. It generated positions as functions of time, which were then compared with the observed sources and labelled as Known Sources in the same way and by the same software module in which bright infrared stars were identified and labelled when their observations appeared in the data stream. This predictor is therefore the first piece of asteroid software, so it is discussed first.

In SDAS a variety of tests were applied to the observations in order to separate fixed sources on the sky from those whose fluxes varied greatly either due to intrinsic

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variation of the source or due to change in apparent position, as would be the case for most solar-system sources. Sources failing this test and having "solar-system-like" colors were written onto magnetic tape for later processing. The package of software which carried out this analysis is known as the Asteroid Tagging Algorithm (ATA). The description of the ATA is given in §4.2 below.

4.1 The SDAS Asteroid Predictor

A description of the "known, source correlation" processing in SDAS is given in the IRAS Explanatory Supplement (ES §V.D.4). This task was performed by a processor named PSCORE for each individual survey observation, which also was the unit of processing in SDAS. The telescope scan control parameters were used to reconstruct the path of the boresight on the sky, and the positions of inertially fixed sources were mapped into the focal-plane coordinate system, in which individual detectors crossed by the image were identified.

The problem was slightly more complicated for solar system objects. In these cases, when the first observation of a SOP was processed, an ephemeris which covered the SOP period was computed for each object in the list of 2500 numbered asteroids, twelve comets, and the outer planets. (Note: This number of asteroids is quite different from the number used later in IMPS, which also did not deal with comets or major planets). Then for each observation, a temporary file was set up which appeared to be just another file of inertially fixed sources to PSCORE. The positions in this file were obtained as follows.

The positions of the moving objects were computed at the beginning and end of the SOP period (about twelve hours) and at sufficient times in between to guarantee an ephemeris for each object with no more than about ten arcminutes separation between snapshots along the path of apparent motion. This was done once for each SOP. Apparent positions were computed in the spacecraft-centered coordinate system. Standard methods were used to compute the ephemerides from osculating orbital elements. Only objects with elliptical orbits were included (in one case, an elliptical approximation to a hyperbolic orbit was used).

As each observation was processed, the positions of all the objects were interpolated to the mid-observation time, and then the time of closest approach of the boresight to the position of each object was interpolated from the scan geometry. The regularity of the scan rate allowed an accuracy of better than three seconds in the predicted scanning time.

Whenever the geometrical calculation indicated a possible intersection of the telescope field of view and the trajectory of one of the moving objects, the approximate crossing time was used to re-interpolate the apparent position of the moving object. This included a first-order light-time correction based on the distance from the spacecraft to the object. The telescope pointing history included aberration corrections for the orbital motion of the earth and the spacecraft. This detailed pointing reconstruction permitted accuracy of better than 0.1 second in the predicted time at which the image crossed each detector. The apparent position of the object was then used as that of an inertially fixed object for the remainder of the computation. In other words, once the file of pseudo-fixed sources was prepared, the rest of the processing did not depend upon the fact that solar-system objects were involved.

After the detector data streams had been processed to extract point-source detections, and after these had been subjected to seconds confirmation and band merging (see ES, §V C and §V D), the known-source predictions were sought among the observations. In order to avoid flux biases, this was done on a position-matching basis only. Associations were tagged by storing the known-source identification codes in the observation data records.

4.2 The Asteroid Tagging Algorithm

The Asteroid Tagging Algorithm (ATA) was a post-processor subroutine of the IRAS SDAS Hours Confirmation Processor (*cf.*, ES). After all normal processing for each seconds-confirmed sighting was completed, the ATA processed the sighting to determine whether it might possibly be a solar system object.

4.2.1 Input Data

The ATA processed the sighting record with which the main hours confirmation processor (PHRCON) had just finished working. At the same time, a field of sightings serving as candidates for hours confirmation remained in a core buffer, and this was also made available to the ATA. In order to be considered usable, a sighting was required either to be seconds-confirmed or at least to have its failure to be seconds-confirmed explainable by the passage of its image over a dead or noisy detector. The parameters defining each detector were as follows:

1950.0 mean ecliptic longitude, latitude, and twist angle,

Position error parameters on the scan and cross-scan axes,

Fluxes and flux error parameters in the four survey channels,

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Detection time and detector identification array,

Status words describing flux quality and confusion level, and

Known-source identifier (zero if no association).

The main routine also provided a set of pointers to hours-confirming candidate sightings, if any.

A. Drop-Dead Sighting

The IRAS SDAS hours confirmation processor operated on a first-in-first-out buffer of sightings spanning 36 hours of survey data. In principle, up to 36 hours could pass before hours-redundant coverage was obtained for any given point on the sky. All data were processed in time order. Hours confirmation was performed for the oldest detection in the 36-Hour File, which was required to be hours-confirmed or to be rejected as soon as the buffer had received data time-tagged at least 36 hours later. For this reason, the sighting being processed was given the name "drop-dead" sighting.

B. Candidate Sighting Buffer

Each drop-dead sighting was accompanied by a collection of other sightings to be used as candidates for hours confirmation of the drop-dead sighting. This collection was formed by gathering together all detections in the 36-Hour File within a spatial window centered on the drop-dead sighting. This window spanned twenty arcminutes of ecliptic longitude and ten arcminutes of ecliptic latitude. The shape was chosen to accommodate the tendency of the position error to be greater in longitude than latitude, and to improve the chances of including subsequent sightings of the same asteroid which produced the drop-dead sighting. The window could not be made larger because of execution time limitations.

4.2.2 SDAS Asteroid Data Output

The asteroid data output from SDAS for processing later by IMPS consisted of positions, fluxes and various status words and other parameters. These were extracted and written to two files as described in the following.

A. CN28 and CN29

The CN28 data file contained a record for each potential asteroid sighting. The CN29 data file was essentially identical, except that it was generated at the weeks-confirmation level, but in fact no additional useful information was contained in this file and it was not used.

B. Statistical Data

Certain statistical parameters were computed by the ATA to aid in tuning the thresholds for asteroid recognition. These included dispersion parameters in position and color, counters for correct and incorrect identifications of known asteroids and known inertially fixed point sources, histograms of asteroid sighting group sizes, and correlation analysis of observed vs. predicted apparent motion in ecliptic longitude and latitude. Many plots of these data are shown in Chapter 6.

4.2.3 Processing

The ATA module was called by the main hours-confirmation routine at the end of the processing for each drop-dead sighting. The ATA processed only seconds-confirmed sightings and candidates. If the drop-dead sighting was hours-confirmed, then the search for associated asteroid sightings was confined to the candidates which the main processor had marked as the confirming sightings; otherwise, all sightings in the coarse processing window were examined by the ATA. This was the only processing step which depended on whether the drop-dead sighting had been hours-confirmed.

A. Main Logic Flow

Each drop-dead sighting was required to pass a color test designed to eliminate objects with non-solar-system colors. If this test was passed, the object was included in file CN28. Whether any other sightings were grouped with it was determined by several additional tests. In order for a candidate sighting to be regarded as a potential sighting of the same asteroid as the drop-dead sighting, it had to pass the same asteroid-color test, followed by a pair of tests which measured its photometric similarity to the drop-dead sighting. The first of these was a flux test, and the second was a color-similarity test. If these tests produced acceptable results, then a motion test was applied to verify that there was at least some probability that the object was moving. Three of the tests required more than a simple yes/no measurement, and the quantity which was computed to decide the issue is referred to as a "figure of merit".

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B. Asteroid Color Test

Each object tested for solar-system color was first classified by a spectral combination code. This indicated the combination of survey channels in which a point-source sighting had occurred. In the case of single-band objects, limits on the colors were obtained from the upper-limit noise estimates in bands adjacent to the detection. Each spectral combination had its own threshold.

The color test involved forming a chi-square parameter as follows. With four survey channels, three independent colors are possible, and so a three-dimensional color space was employed. The locus of points corresponding to solar-system objects was modelled as a straight line joining the points provided by a standard thermal model (see Chapter 7) for an asteroid at 0.87 AU and one at 40 AU. Sightings with three colors were treated as a point in this space; sightings with two colors were treated as points in the corresponding plane, and the projection of the model line into that plane was used as the nominal locus of points for solar-system objects. Sightings with only one color were treated as points on the corresponding axis, and the projection of the model line onto that axis was used as the range into which the point should fall. In practice, the only single-color sightings were those observed only in the 12 μm channel or the 100 μm channel, since two color estimates could be derived from the noise-fill upper limits for the other single-band sightings. For example, a sighting detected only in the 25 μm band had detector noise data for the 12 μm and 60 μm bands from which two low-quality color bounds could be obtained. Of course, these had larger uncertainties than colors derived from full-fledged detections, but they often sufficed to rule out spurious data.

Once the appropriate color space was determined, the minimum distance from the observed point to the nominal locus was computed and treated as an error vector. The squared components of this error vector served as the numerators of the chi-square terms; the corresponding denominators were the *a priori* uncertainties in the observed colors. This produced a chi-square parameter with one, two, or three degrees of freedom, depending on the number of independent colors available. The figure of merit was taken to be the fraction of all chi-square random variables with the same number of degrees of freedom and larger magnitudes. Thus the figure of merit ran from nearly zero for very large error vectors to nearly unity for very small error vectors.

The computation of the minimum distance from the observed color point to the nominal locus took into account the finite extent of the locus. In other words, the nominal locus was a finite line segment; the error vector was first computed as the perpendicular displacement from the line containing this segment to the observed color point; if the error vector intersected the line outside of the range of the nominal

locus, then a non-perpendicular line from the nearest end of the nominal-locus range to the observed color point was used instead. In the case of single-color observations, the test degenerated to whether the point was contained within the locus projection on the axis; if so, the error vector was null, and otherwise the distance from the observed point to the nearest end of the locus range was used as a one-dimensional error vector.

The figure of merit was required to be above a threshold set for the specific spectral combination of the sighting. If the drop-dead failed this test, the ATA ceased processing it, proceeding to the statistical computations described below. Candidates which failed the test were dropped from consideration.

Note that while the four IRAS survey channels yield three independent colors in the sense that there are three degrees of freedom, the colors are not all statistically independent. The error in the color derived from the first and second bands is correlated with the error in the color derived from the second and third bands, because the error in the second band affects both colors. The correlation coefficient for the random errors in these two colors is 0.5. Ignoring the correlation resulted in underestimating the variance of the corresponding chi-square parameter by 20%. In the case of three-color tests, the variance was underestimated by 25%. While these correspond only to errors of 11% and 13% in the standard deviation, and hence are about equal to the estimation error in the *a priori* photometric random error, the thresholds for the corresponding spectral combinations were nevertheless set lower than for other combinations in order to compensate for the approximation error.

C. Common Flux Test

The common flux test employed all bands in which the drop-dead and candidate sightings were both detected; if there were none, the test was skipped. The test simply required the fluxes in such bands to be within a factor of ten of each other. This accepted more than the highest known light-curve variation of the time scale of a few hours, while eliminating the most obvious mismatches.

D. Common Color Test

The colors of each sighting in the pair to be tested were computed as described above in §4.2.3 B *i.e.*, with color bounds based on detector noise used for bands in which no detection occurred). Only colors common to both sightings were used, and if there were none, the test was failed. Otherwise a chi-square random variable was computed by summing terms with numerators equal to the squared difference in a common color and denominators equal to the sum of each sighting's error variance for

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the corresponding color. The number of degrees of freedom was the number of terms summed, and the figure of merit was the cumulative chi-square distribution for that number of degrees of freedom evaluated at the observed value of chi-square. This figure of merit was required to be above a threshold which depended upon the number of bands detected. The corresponding fractions of all true sightings pairs intended to be retained ranged from 0.99 to 0.998.

E. Angular Motion Tests

An apparent angular motion test was performed which examined pairings of the drop-dead sighting with each hours-confirmed candidate, if any. This test was not required if the drop-dead sighting was not hours-confirmed. The test employed the mutual position information of the drop-dead sighting and the candidate to determine whether it was consistent with the hypothesis that the object was moving. The figure of merit was an approximation to the probability that the true position of the drop-dead and candidate sightings were not contained within the same small region of sky corresponding to a square approximately 2.4 arcminutes on a side. This probability was required to be above 0.001. This eliminated only sightings that were virtually certain to be inertially fixed, i.e., sightings with very low position uncertainties and very good position agreement. Hours-confirmed sightings with medium-to-low confirmation scores passed this test easily.

F. Known Object Analysis

After completing all decisions concerning whether sightings were moving objects, a check of the known-source identifier in the drop-dead sighting's parameter record was made; if any accepted candidates had been found, these were also checked. Values of zero indicated that the sighting was not a known object; values between one and 30,000 indicated that the sighting had been associated with an inertially fixed object; values above 30,000 indicated association with a solar-system object.

In most cases, all sightings either were not associated with any known object, or else they were all associated with the same known object. Separate counters were maintained for the combinations listed in §4.2.3 G.

G. ATA Sighting Association Counters

Rejected drop-dead sightings

1. Not a known object
2. Known solar-system object
3. Known inertially fixed object

Accepted drop-dead sightings

4. No sightings of known objects
5. Some sightings of known objects
6. All known objects were solar-system objects
7. All known objects were inertially fixed objects
8. Some solar-system and some inertially fixed objects

The statistical parameters discussed in the next section were maintained for each of the combinations above. When known solar-system objects were found, they were output to CN28 even if they had failed to be accepted by the ATA. When known inertially-fixed objects were found, they were not output to CN28 even if they had been accepted by the ATA. The separate statistical analysis for each combination was used to tune the threshold parameters.

H. Statistical Data Gathering

All statistical counters were broken down into ten groups based on the highest signal-to-noise ratio in the set of sightings. In addition, a breakdown by the various combination of known-object identifiers was performed. For known asteroids, correlation coefficients were computed from observed and known motion rates in latitude and longitude. Excellent correlation in latitude was obtained, but as expected, the lower resolution in longitude yielded marginal statistical significance.

4.3 The IRAS Minor Planet Survey (IMPS)

The scientific processing and analysis of the asteroid data was the chief goal of the IRAS Minor Planet Survey (IMPS). As its input, IMPS took the file of candidate sources originally compiled by SDAS, as well as other files of asteroid information supplied by E. Tedesco (*cf.*, Chapter 3). The chief output from IMPS was the series of data products presented and described in Part II. This chapter presents a technical documentation of IMPS. The description is organized in the order of input, output and then the details of the various processors and their algorithms.

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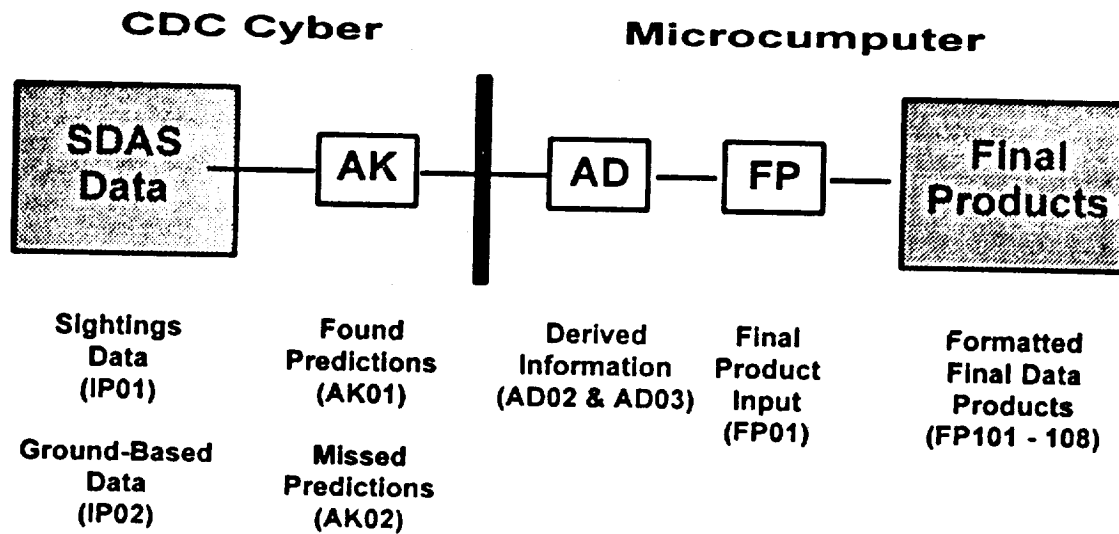


Figure 1. Schematic diagram of IMPS data flow. Filled boxes indicate input and output, open boxes the major subsystems used in processing potential asteroid sightings. Intermediate files produced are indicated below each subsystem. The heavy vertical line indicates the interface between the computing platforms used to perform the processing.

Before getting into the details, it is useful to orient oneself by studying a block diagram of the system presented in Fig. 1. Data input and output are shown as shaded boxes. The other boxes show the various processors and other parts of the system. The reader will note that IMPS is split across two hardware platforms, the CDC Cyber mainframe and the PC workstation. The mainframe was needed to hold the large amount of input data to IMPS (comprising about 700 megabytes) as well as the AK processor, which, for the large number of asteroids that IMPS processed, is a very CPU- and I/O-intensive program. Once AK produced its output files, these files could then be transferred to the PC workstation where subsequent processing and user interface tools could be implemented.

The AK processor handles position association for known asteroids (*i.e.*, Asteroids, Known = AK). AD is where radiometric diameters and albedos are calculated (*i.e.*, Asteroids, Derived parameters). FP stands for Final Products. FP is essentially a threshold and format processor. Here the final acceptance parameters for accepted sightings can be set or reset just prior to producing the Final Data Products. Some of the processors have subprocessors, and their names always have the letters of the parent processor as the first two letters.

Most of the work of the processors is accomplished by setting flags or by filling in blank parameters with meaningful values. All of the sightings remain available in the parent data base. The data base as actually implemented contains a variety of files not shown in the high-level schematic of Fig. 1, and these will be discussed later as needed.

The primary goal of IMPS is to obtain derived information concerning asteroids with reliable orbital elements. The first implementation task was to set up the input data bases; this comprises the shaded box in Fig. 1 labeled "SDAS Data". The main data bases are those known as IP01 and IP02, the IRAS sightings and the ground-based asteroid data bases, respectively. The former is primarily composed of the ATA output, but space was included for key IMPS parameters to be added. The latter is similar to the numbered-asteroid data base used by SDAS, but is considerably expanded in terms of the number of objects involved and the physical parameters included; three sets of orbital parameters were provided for each asteroid to better define its position in space for the period during which the IRAS satellite was operational.

The second task was to tag sightings in the IP01 data base with predicted sightings of known asteroids. This yielded, among other parameters, figures of merit regarding the positional coincidence of sightings to solar-system objects; thresholds for this quantity governed acceptance and rejection of sightings at later stages.

The third task was to derive albedos and diameters for the known asteroids which were found in the IP01 sightings file. This involved computing these parameters for each IRAS detection of the object in any survey band, and then averaging them to obtain the best overall estimates for each object. In the process of averaging the data, considerable editing was performed to remove suspected spurious results caused by the various IRAS noise sources. Some asteroids had no usable data remaining after this editing.

The fourth task was to gather all of the information together into the formats needed for deliverable products (machine-readable data bases and hard copy) and to produce documentation describing these products and the steps taken to obtain them.

4.3.1 Input Data

Input came from a variety of sources, some generated by machine, others compiled by hand. There were seven machine-readable input files used. A complete description of the ground-based files used is found in Chapter 11 of this volume. A summary of the IMPS input files include:

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IP01 - probable asteroid sightings (~2.7 million)
IP02 - known asteroid orbital elements
PR04 - IRAS survey observation parameters
SHEF - Spacecraft Heliocentric Ephemeris
Thermod - asteroid thermal model
XYZSun - solar ephemeris
AACV - IRAS boresight area coverage

4.3.2 Output Data

The output from IMPS consisted of a number of asteroid data products. These are described in Chapter 9, the formats of the machine-readable files are given in Chapter 10, and the hard-copy versions are given in Chapters 12 through 16.

4.3.3 Data System Flow

The input data bases were set up as previously described in §4.3.1. The next data set produced was the file AK01, which contained pointers and association information linking objects (*i.e.*, numbered asteroids) with sightings in IP01. AK01 provided the mechanism for accessing sighting data for any given object in all downstream processors. Predicted sightings which were not realized were recorded in AK02; this included sightings which were actually impossible because, for example, the source was too faint or the image crossed dead detectors, etc. Positional associations of AK01 records with IRAS Catalog objects, Faint Source Survey objects, and members of other astronomical catalogs were made off-line and recorded in the appropriate status word bit (in AStatW) in IP01.

At this point, ASCII-formatted files containing pertinent information from IP01, IP02, AK01, and AK02 were down-loaded to the PC workstation for all subsequent processing.

Albedos and diameters for the solar-system objects, computed by the AD subsystem for each detection of the known object in any survey band, were output to files AD02 and AD03. Upper limit derivations of albedos and diameters were output to file AD04. Processing log files were also maintained.

The FP subsystem performed the editing of the AD02 and AD03 data and the averaging of the remaining estimates for each source. This produced the intermediate files FP01A and FP01B for asteroids types 1 and 2, respectively. Then another FP module gathered all of the various parameters together from all the different files and produced the machine-readable versions of the final products for known objects.

4.3.4 System Processing Steps**A. Input Data Preprocessing**

Input data preprocessing was performed by the IN subsystem. This involved the following tasks:

- a. Conversion of CN28 (ATA output) to IP01
- b. Association of IP01 sightings with CN29 records
- c. Setting of high-density-region flags in IP01 records
- d. Setting of faint-asteroid flags in IP01
- e. Conversion of the SDAS Area Coverage File to the ADAS Area Coverage File
- f. Uploading and formatting of the Known Object File (IP02)
- g. Conversion of the SDAS Spacecraft Heliocentric Ephemeris File to the ADAS Spacecraft Heliocentric Ephemeris File
- h. Apply SDAS final calibrations (including hysteresis corrections) to IP01 fluxes
- i. Setting of outer-slot-only flags for IP01 sightings whose detectors all bordered the cross-scan survey array limits.

B. Known Object Prediction

Each object in the IP02 data set was processed by the AK subsystem to search for matching IRAS sightings. The entire mission was searched for each object before going on to the next. The process for a single object was performed in two phases: first geometrical coincidence of the object's trajectory with that of the scanning telescope was sought; then all such coincidence times were checked for actual sightings occurring sufficiently close in position and time. The first phase was performed by a module named AKSOPS, and the second phase was performed by a module named AKSITS. The algorithms used by these modules will be described next.

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The AKSOPS Module

AKSOPS first determined all of the Area Coverage File (AACV) bins that the asteroid traversed during the entire survey mission. The AACV bins are records in a direct-access data set; each bin corresponds to a particular area of the sky, and the contents of the record are times of entry and exit of the telescope boresight with respect to that area of sky. The bin numbers were buffered in an array, and duplications were eliminated. Then each bin was examined to see when the IRAS telescope boresight traversed it. Whenever a telescope entry and exit time pair was found to overlap the time during which the asteroid was in the bin, the asteroid position was recomputed for the midpoint time of telescope boresight passage. The boresight typically spent from a few seconds to a few minutes in any given bin. With the refined asteroid position, the PR04 data for the observation involved was checked, and the solar aspect angles of the asteroid and the boresight were compared. If they were within sufficient proximity of each other, the trajectory-crossing parameters were prepared for AKSITS, and the crossing counter was incremented.

Standard techniques were used to compute positions as functions of time for the osculating orbits about the sun and for transforming these into the 1950.0 mean ecliptic coordinate system. The vector from the spacecraft to the asteroid (expressed in the same system) was obtained by subtracting the vector from the sun to the spacecraft from the heliocentric asteroid vector; the spacecraft vector was obtained by interpolation from the Spacecraft Heliocentric Ephemeris File (SHEF).

The position of the asteroid was computed for a variety of times as a search for all of the AACV bins it traversed was carried out. These times were chosen in the following way. First the mission start time was taken as the epoch of interest. Each asteroid has three sets of orbital elements corresponding to times distributed over the IRAS mission. The orbital element set whose time tag was closest to the epoch being processed was selected each time a position was to be computed. The SHEF file was read up to the first epoch, and the counter for the number of asteroid-telescope crossings was initialized to zero. The asteroid position for the time point was computed, and the time needed to travel ten arcminutes (as seen from the spacecraft) was computed. This time interval is denoted DELTIM, and it is obtained as described below. This distance was chosen to control the sampling of the asteroid position because it eliminated the possibility of skipping over a bin crossing by the asteroid.

In the orbital plane, standard (x,y) coordinates are defined by

$$x = a \cos E \quad (1)$$

and

$$y = b \sin E \quad (2)$$

where a is the semimajor axis, b is the semiminor axis, and E is the eccentric anomaly (obtained by solving Kepler's equation for the time point being processed). The origin of this coordinate system is not the sun, but that does not matter since we will be using only the derivatives. Note also that these (x,y) coordinates are not consistent with the IRAS conventions, and this will have to be reconciled below. We have

$$dx = -a \sin E \, dE \quad (3)$$

$$dy = b \cos E \, dE \quad (4)$$

The energy integral of motion for the two-body problem provides the relation

$$v^2 = k^2 M_s \left(\frac{2}{r} - \frac{1}{a} \right) \quad (5)$$

where k is Gauss's gravitational constant, and M_s is the mass of the sun. This last equation yields the magnitude of the velocity vector, and the two preceding it can be used to obtain the direction. Noting that

$$\frac{b}{a} = \sqrt{1 - e^2} \quad (6)$$

we can construct a vector (w_x, w_y) which is parallel to the velocity vector $(dx/dt, dy/dt)$, or

$$w_x = -\sin E \quad (7)$$

$$w_y = \sqrt{1 - e^2} \cos E \quad (8)$$

This can be unitized by dividing by the root-sum-square magnitude w , which can be reduced to

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$$w = \sqrt{1 - (e \cos E)^2} \quad (9)$$

This leads to the velocity vector components

$$v_x = -\frac{v}{w} \sin E \quad (10)$$

$$v_y = \frac{v}{w} \sqrt{1 - e^2} \cos E \quad (11)$$

We will now relabel the axes so that they are consistent with the IRAS conventions used elsewhere throughout this document. This involves identifying the x component as the IRAS z component, the y component as minus the IRAS y component, and setting the IRAS x component to zero, so that

$$v_x = 0 \quad (12)$$

$$v_y = -\frac{v}{w} \sqrt{1 - e^2} \cos E \quad (13)$$

$$v_z = -\frac{v}{w} \sin E \quad (14)$$

This vector must be rotated about the x-axis in order to align the z-axis with the line of nodes; then it can be transformed into the IRAS (1950.0) mean ecliptic coordinates by the standard Euler rotations. The velocity of the earth can be approximated to sufficient accuracy for these purposes by assuming its mean magnitude, 29.786 km/s, and a direction given by the cross product of the ecliptic north-pole unit vector with the unit vector from the sun to the spacecraft. This is subtracted from the asteroid's velocity vector; we will denote the result V_v and the vector from the spacecraft to the asteroid will be denoted V_s . Then the angle θ between V_v and V_s is

$$\theta = \cos^{-1} \frac{V_v \cdot V_s}{|V_v| |V_s|} \quad (15)$$

where $|V_v|$ is the magnitude of V_v and r_s is the magnitude of V_s . The component of the velocity vector perpendicular to V_s is therefore $v \sin \theta$, and since this is viewed from a distance of r_s , the angular velocity of the asteroid as seen from the spacecraft, μ_s , is

$$\mu_s = \frac{v \sin \theta}{r_s} \quad (16)$$

The time required to traverse approximately ten arcminutes (actually, three milliradians) is

$$\Delta T = \frac{0.003}{\mu_s} \quad (17)$$

The second time point follows the first by the interval ΔT . At this new epoch, the position and velocity were recomputed, along with ΔT ; in this way, the time required to traverse about ten arcminutes was continually re-estimated, and this permitted the set of time points to be generated.

At each time point, the asteroid position was obtained. A search square was centered on the asteroid position such that the sides of the square were aligned with the local lines of longitude and cross-longitude of the L0 (1950.0 mean ecliptic) system, with a half-width of five milliradians. The four corners of this square defined points in latitude and longitude for which the AACV bins numbers were obtained via a standard mapping algorithm. All bin numbers were buffered in an array, with duplicates removed.

The SHEF data were read and interpolated for each time point. The SOP granularity of the time domain affected the generation of time points in a way which modified the description above slightly. This involves constraining the beginning and ending times of each SOP to be used as time points, so that brief visitations in certain bins will not be overlooked.

After all bin numbers were identified for a given SOP, the following processing was performed before going on to the next SOP. The Area Coverage File data for each bin number were read in, and for each boresight passage in the bin, the mean of the entry and exit times was computed; the asteroid position at this time was computed as described previously. The SHEF data were interpolated to the time being processed,

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and topocentric correction was performed. This yielded the vector from the spacecraft to the asteroid; this was not yet corrected for one-way light time delay, but this correction was usually much less than 30 arcseconds. While this is still more than the topocentric correction, the approximate asteroid position was used only for a relatively coarse search (the light-time delay was accounted for later in AKSITS). The spacecraft position vector was sufficiently accurate to be used again in the more detailed asteroid position computation in AKSITS, and so it was retained in a buffer for that purpose.

The PR04 data for the time being processed were read in next, and the mean value of the boresight Euler angle ν was found, $\langle \nu \rangle$; the corresponding angle for the asteroid, ν_a , was computed as follows.

$$\nu_a = \cos^{-1}(\cos \beta \cos(\lambda - \lambda_s) - \frac{\pi}{2}) \quad (18)$$

where λ_s is the solar longitude at the middle of the OBS, and λ and β are the longitude and latitude of the asteroid in the L0 system (1950.0 mean ecliptic), corresponding to the vector computed by subtracting the sun-to-spacecraft vector from the sun-to-asteroid vector. This vector is denoted V_a and is defined in the L0 system, which has its Z axis in the direction of the north ecliptic pole and its X axis in the direction of the vernal equinox; therefore

$$\lambda = \tan^{-1} \frac{-V_{a2}}{V_{a3}} \quad (19)$$

$$\beta = \tan^{-1} \frac{V_{a1}}{\sqrt{V_{a2}^2 + V_{a3}^2}} \quad (20)$$

If the absolute value of $(\nu_a - \langle \nu \rangle)$ was less than 15 arcminutes + $3 \sigma_v$, then geometrical coverage was considered to be detected, a crossing counter (ICRS) was incremented by one, and the crossing parameters were loaded into a buffer array for later use by AKSITS. The estimated detection time T_{ICRS} was computed as follows: the asteroid's angle about the sun vector, ψ_a , is

$$\psi_a = \tan^{-1} \frac{\cos \beta \sin(\lambda_a - \lambda)}{\sin \beta} \quad (21)$$

If this was negative, 2π was added; this made ψ_a conform to the range of the boresight's angle about the sun vector. The time it would take the boresight to get to the asteroid from its position ψ_0 at time T_0 is $(\psi_a - \psi_0)/\psi'$, where ψ' is the mean scan rate of the angle about the sun vector during the scan; adding this scanning time to T_0 yielded the estimated time of closest approach of the boresight to the asteroid. Since detection should occur a few seconds before this, two seconds were subtracted, and the result was stored in T_{ICRS} .

When all of the bins identified for possible geometrical crossing had been prepared, the next SOP was processed as described above until the entire mission had been covered for the given object. At that point the value of the crossing counter, ICRS, provided the number of predicted crossings of the asteroid by the IRAS focal plane.

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AKSITS processed the geometrical crossings one at a time; for each one, the following processing steps were executed.

A time window was set up by adding and subtracting ten seconds to the T_{ICRS} for the crossing being processed. Then the one-way light-time correction was computed by dividing the distance to the asteroid from the spacecraft by the speed of light. This correction was subtracted from the time for which AKSOPS computed the asteroid position, and this position was recomputed. The vector from the spacecraft to the asteroid was computed by subtracting the sun-to-spacecraft vector from the sun-to-asteroid vector, where the former is the one for the expected instant of observation, and the latter is for the time that the light left the asteroid. The asteroid position computation was performed as described above.

When the asteroid's LO position angles (as seen from the spacecraft) had been computed, a time window of IRAS sources was examined to see how well each position matched that of the computed asteroid position.

Position agreement was tested as follows. A coarse latitude test was applied first; this eliminated most of the sightings in the time window. If the latitude test was satisfied, a dot product test was performed. Unit vectors toward the computed asteroid position and the sighting were dotted, and if the result was less than 0.9999 (i.e., the angular

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separation was 0.8° or more), the sighting was discarded; otherwise the fine position test was performed.

The fine position test is the same one used by the SDAS PSCORE processor (see §4.1). It computed the cross-covariance of the two position probability density functions (i.e., for the sighting and the computed asteroid position) evaluated at the observed separation, and required the result to be above the threshold 1000 (the density functions had units of probability mass per steradian). If this test failed, the sighting was discarded; otherwise it was checked against any previous acceptable results which may have been found. If there were any, then bit number 31 in the AStatW status word was turned on to flag possible confusion, and the association with the higher test result was retained.

The units employed in the position probability density functions resulted in the figure of merit having a large value (1,000 up to as much as one billion). The corresponding figure of merit used subsequently in IMPS (for such things as final product filtering) was obtained by taking the common logarithm of this figure of merit, subtracting three, and dividing by six. This resulted in a number between zero and one.

A flux ratio test was applied to all sighting/prediction pairs which passed the fine position test and for which the sighting had a signal-to-noise ratio of at least 5.0 at $12\ \mu\text{m}$ or $25\ \mu\text{m}$. In such cases, if the sighting's flux was not within a factor of three of the flux predicted by the standard thermal model in either band, then this failure was flagged by setting AStatW bit number 7. If any other sightings also passed the fine position test, the best of the unflagged ones was kept, if any, even if the position score was less than that of a flagged sighting.

After the time window of sightings was exhausted, AKSITS noted whether any associations had been made. If not, then a missed-prediction record was written to the AK02 file. If an association was found for the current prediction, then a check was made for an existing AK01 record which would indicate that the sighting involved had already been used in a previous match with another known object. In such a case, AStatW bit number 30 was set, and if one and only one of the two was flagged as having failed the flux test, the other was kept, and otherwise the association with the higher test result was retained.

If there was no conflicting match, or if there was and the current association was preferred, then AKSITS generated an AK01 record to record the association for the given crossing.

After each predicted sighting for the current asteroid had been processed, AKSITS went on to the next predicted sighting until all had been processed.

C. Derived Information Processing

Albedos and diameters were computed for each known object by applying the same algorithm to each detection in any survey band. The results were averaged for each object later during final product preparation, at which time additional editing criteria were applied to weed out unreliable data. The computation of albedo for each detection employed a table of normalized fluxes as a function of Bond albedo and heliocentric distance; this table (*cf.*, Appendix 4) was provided by L. Lebofsky and was derived from the IRAS standard thermal model (*cf.*, Lebofsky *et al.*, 1986a,b). The remaining discussion in this section will be concerned with the computation of the albedo for a given detection of a given asteroid.

It is well known that source detection via thresholding on flux or signal-to-noise ratio induces a flux overestimation near the threshold. Corrections for this effect were derived empirically from measurements of asteroids observed by both the IRAS instrument and the Infrared Telescope Facility (IRTF) at Mauna Kea (See §8.3.2 for complete details). The correction was applied to all detected fluxes with signal-to-noise ratios (SNR) between 3.0 and 10.0. The correction factor was interpolated linearly between these values, with the correction factor being 0.725 at SNR = 3.0 and 1.0 at SNR = 10.0. This was done before iteratively computing the albedo and diameter. The flux uncertainty for corrected bands was increased by root-sum-squaring it with the flux correction.

In addition, another correction was applied after the albedo had been obtained; a correction factor of 1.12 was applied to albedo solutions using data from 25, 60, and/or 100 μm . This was done in order to eliminate an observed bias between these bands and the 12 μm band. (See §8.3.3 for further details).

An initial estimate for the geometric albedo p_H was taken by assuming the IP02 value. Then the following iteration was performed. After the first band was processed, any subsequent bands began iteration with the previous albedo solution.

- 1.) The phase integral Q was obtained from integration over the IAU Commission 15 visual (V) wavelength phase function,

$$Q = 0.29 + 0.684 G \quad (22)$$

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where G is the slope parameter.

- 2.) The Bond albedo was computed from the phase integral and the current estimate for the geometric albedo,

$$BOND = Qp_H \quad (23)$$

The Bond albedo was used for looking up and interpolating fluxes in the thermal-model table.

- 3.) The current estimate for the radius was obtained from

$$R = 0.5 \times 10^{(3.1236 - 0.2H - 0.5 \log p_H)} \quad (24)$$

- 4.) The flux was interpolated in the thermal model table for the current albedo and heliocentric distance. The interpolation was linear in albedo and quadratic in heliocentric distance; the flux was then scaled for the current estimate of the radius and the distance from the spacecraft to the asteroid, and the phase-angle correction was applied by the formula

$$F = 10^{\frac{-0.572958 |\alpha_p|}{2.5}} \quad (25)$$

where α_p is the phase angle in radians; the interpolated flux corrected for the radius and viewing distance was multiplied by F .

- 5.) The ratio of the computed flux to the observed flux was calculated, and the absolute deviation from unity was tested for convergence. If it was less than 0.00001, then the albedo was considered converged, and for bands 2, 3, and 4 (i.e., 25 μm , 60 μm , and 100 μm) the correction factor of 1.12 was applied and the radius was recomputed. If convergence was not achieved, but the number of iterations had reached 100, then iteration ceased, and the failure to converge was flagged by setting bit number 1 in the ADSTAT status word for the band. Otherwise the geometric albedo p_H was scaled by the ratio of the computed flux to the observed flux, and the iteration resumed at step (1.) above.

The uncertainties, σ_{PH} and σ_R in the geometric albedo and radius, respectively, were computed as follows. The radius was treated similarly to the albedo, so only the latter will be discussed. The iteration described above was repeated with the one-sigma IRAS flux uncertainty added to the observed flux. The solution for the albedo in this case was subtracted from the unperturbed value for that band, and the absolute difference was taken as the one-sigma uncertainty due to the error in the IRAS flux.

D. Data Base Management

Nearly all of the computer files used during IMPS production on the CDC mainframe were defined as keyed segment access files to greatly decrease computer I/O overhead in dealing with the large files defined in §4.3.1. A user-defined access key is used to point directly to the records desired in the data base. The data access key used was the TNAM/DNAM of an individual sighting, that is, the first time and telescope detector ID of initial detection, which is guaranteed to be a unique sighting identifier.

Given this unique data access key, information for a particular sighting could be easily retrieved from IP01, as well as prediction information from AK01. Most of the data base access routines were primitive subroutine calls to read the various files whose retrieved record's data would be stored in a FORTRAN COMMON block used by the calling program.

E. Special Processing

The Special Processing (SP) subsystem performed a variety of miscellaneous services. These included computing the geometrical coverage completeness for specific orbital element sets (see Chapter 8), and generating a large set of ad hoc intermediate products to support quality checking and the setting of thresholds for final product preparation (see Chapters 5 and 6).

F. Final Product Preparation

Albedo averaging was performed for each known object by forming a set of albedos composed of those determined from each detection in any survey band, editing this set according to certain rules, and averaging the remaining values. The values averaged were also tested for the presence of significant flux variations. Final diameters were computed directly from the final albedos.

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The initial editing involved careful inspection of each sighting's status words, and certain conditions eliminated the sighting from further consideration. The following conditions eliminated entire sightings.

- 1.) The sighting was associated with a source in the IRAS Working Survey Data Base (WSDb), Point Source Catalog (Ver. 2), Faint Source Survey Catalog (Ver. 2), or Serendipitous Survey Catalog.
- 2.) All detections occurred on detectors at the edge of the survey array.
- 3.) More than one IRAS sighting passed the position-match test with the asteroid's predicted position.
- 4.) The cross-scan position uniform uncertainty component exceeded 5 arcminutes.
- 5.) The normalized position-match figure of merit was less than 0.4.
- 6.) A color test was failed by a multi-band sighting; the parameter tested was the natural logarithm of the ratio of the flux at 12 μm to that at 25 μm , or the similar ratio for 25 μm to 60 μm , or both, if available; the first was required to lie in the range from -0.75 to +1.0; the second was required to lie in the range from 0.5 to 2.25.
- 7.) A sighting with only one band containing a detection of an asteroid for which no other acceptable sightings were found had a flux status of less than 5 (fully seconds-confirmed).

Within each sighting, some bands may have been able to contribute to the overall average and others not. An observation in a given band was excluded if any of the following conditions were found.

- 1.) The observation occurred in band 4 (100 μm).
- 2.) The flux status in band-2-only (25 μm) sightings was less than 4 (non-seconds-confirmed but possibly due to a failed or noisy detector).
- 3.) The confusion status word indicated possible confusion at some confirmation level.
- 4.) The detection correlation coefficient was below threshold (0.94 for 25 μm and 0.90 in the other bands).

- 5.) The albedo iteration did not converge, or the value fell outside the range 0.01 to 0.90.
- 6.) The albedo was too far from the mean of the other albedos (Chauvenet's criterion was used for this purpose).

A more detailed discussion of the data filtering and how it is believed to have affected completeness and reliability is contained in Chapter 6.

When all observations had been identified as usable or unusable, the averaging procedures were carried out. The sample mean for the set of usable measurements was obtained by summing the albedo and the square of the albedo over all usable measurements, while counting them. The mean was computed as an unweighted arithmetic mean at this point, since the samples could come from different populations (including noise), making inverse-variance weights untrustworthy. The unbiased estimate of the population variance is

$$\sigma_{pHp}^2 = \frac{N}{N-1} \left(\frac{\sum p_H^2}{N} - \left(\frac{\sum p_H}{N} \right)^2 \right) \quad (26)$$

where the summations are over the albedos remaining after filtering out ineligible results, and N is the number of measurements contributing to these sums.

The overall average was recomputed by disqualifying observations for which the albedo was different from the mean by more than allowed by Chauvenet's criterion (which chooses a limit in units of the standard deviation σ_{pHs} appropriate for the number of samples). Again, an unweighted mean was computed, and an estimate of the uncertainty in the mean was obtained by dividing the root-sum-squared *a priori* measurement errors by the number of samples averaged. This variance is denoted σ_{pHs}^2 . The mean value for the albedo was the final answer for that parameter, and is denoted $\langle p_H \rangle$.

Once the final value for the albedo was known, all the usable sightings were examined once again to see if any had all of their albedo estimates above or all below this final value. This was done to investigate whether light-curve (or other systematic) effects were present in the observations. When such sightings were found, the in-sighting mean albedo $\langle p_{Hs} \rangle$ and its reduced variance σ_{pHs}^2 contributed a term of the form

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$$\frac{(\langle p_H \rangle - \langle p_{Hs} \rangle)^2}{\sigma_{p_{Hs}}^2} \quad (27)$$

to a summation of such terms over all such sightings. This summation constitutes a sample of a chi-square random variable (under the hypothesis that the observations were all made under identical conditions with the same true value $\langle p_H \rangle$ being measured with associated measurement error) with N degrees of freedom, where N is the number of sightings contributing to the summation.

The probability $P(\chi^2)$ associated with this chi-square value is the fraction of all chi-square random variables with the same number of degrees of freedom which have the same value as the observed sample or less. If the observed chi-square value was rather large, $P(\chi^2)$ would be near unity; this should happen if statistically significant sighting-to-sighting discrepancies in the albedo occur, and a noticeable light curve should emerge in this way (other effects could also cause it, however). If the chi-square value was mediocre, then $P(\chi^2)$ would be approximately 0.5. If the chi-square value was rather small, it probably meant that the *a priori* uncertainties were overestimated, and the value of $P(\chi^2)$ would lie between zero and 0.5.

Since the effect of a significant light curve would be to put $P(\chi^2)$ between 0.5 and unity, a variable F_{LC} (parameter PLC in final products number 102 and 103) was derived from

$$F_{LC} = 2 P(\chi^2) - 1 \quad (28)$$

$$\text{If } F_{LC} < 0.1 \text{ then } F_{LC} = 0.1 \quad (29)$$

and we loosely refer to F_{LC} as the probability that a light curve effect is present in the data. F_{LC} was then used to set the uncertainty of $\langle p_V \rangle$ according to

$$\sigma_{\langle p_V \rangle} = \sqrt{F_{LC} \sigma_{p_{Hp}}^2 + (1 - F_{LC}) \sigma_{p_H}^2} \quad (30)$$

The value of F_{LC} was not permitted to go below 0.1 as a safety measure. The equation immediately above causes the uncertainty in $\langle p_V \rangle$ to approach the reduced standard deviation σ_{p_H} if it appears that the measurements are all quite consistent with each other (which would rule out significant light curve effects), and to approach the

population standard deviation σ_{pap} in the limit that the measurements are significantly inconsistent with each other.

Finally, the diameter was computed from $\langle p_H \rangle$ via

$$D = 10^{(3.1236 - 0.2H - 0.5 \log \langle p_H \rangle)} \quad (31)$$

The uncertainty in the diameter is set so that its relative uncertainty is half that of the albedo. The AStatW status word bits 12+i are set on if band i (i = 1, 2, 3) contributed to the final average albedo; this is done for each sighting.

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